ALGEBRA 1 · EXPONENTIAL FUNCTIONS

Linear vs. Exponential Race

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Linear vs. Exponential Race** on Mr. K's Classroom: Algebra 1 → Exponential Functions → Interactive Math. Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3, then do Part 4 on your own.

1 Predict before you touch the tool

1. Starting at 2, which grows faster in the long run: adding 10 each step or doubling each step?

   Doubling (exponential).

2. What do you think happens to $y = 2 \cdot b^x$ when b is between 0 and 1?

   It decreases: exponential decay.

2 Explore use the tool and record what you see

Race: $y = 4x + 2$ and $y = 2 \cdot 2^x$

Set a = 2, b = 2, m = 4 and c = 2.

x	Linear y	Change	Exponential y	Change
0	2	—	2	—
1	**6**	**+4**	**4**	**×2**
2	**10**	**+4**	**8**	**×2**
3	**14**	**+4**	**16**	**×2**
4	**18**	**+4**	**32**	**×2**
5	**22**	**+4**	**64**	**×2**

Change a and b in $y = a \cdot b^x$

a	b	y-intercept	Growth or decay?	Percent change per step
3	2	**(0, 3)**	**growth**	**100% more**
5	0.5	**(0, 5)**	**decay**	**50% less**
4	1.25	**(0, 4)**	**growth**	**25% more**
2	1	**(0, 2)**	**neither (flat)**	**0%**

3 Write the rule fill in each blank from what you observed

1. A linear function changes by the same **amount** each step. An exponential function changes by the same **factor** each step.
2. In $y = a \cdot b^x$, a is the **starting value (y-intercept)**.
3. If $b > 1$ the function shows **growth**. If $0 < b < 1$ it shows **decay**.
4. A decay graph gets closer and closer to **y = 0** but never reaches it.

5. An exponential growth function eventually **passes** any linear function.

4 Check no tool: show your work

1. Linear or exponential: 5, 15, 45, 135?

   Exponential (×3)

2. Linear or exponential: 7, 11, 15, 19?

   Linear (+4)

3. Describe $y = 100(0.8)^x$.

   Starts at 100 and decays by 20% each step

4. A town of 500 grows 10% per year. Write a model.

   $y = 500(1.1)^x$

5 Reflect

How can you tell from a table whether a function is linear or exponential?

Check the change between steps. Equal differences mean linear; equal ratios mean exponential.